

1D ising model

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$$H = - \sum_{ij} J_{ij} s_i s_j - \tilde{\mu} \sum_i H_i s_i + \mathcal{O}(s_i s_j s_k) \quad (1)$$

We discussed the homogeneous case

$$J_{ij} = \begin{cases} J_{i,i+1} = J & \text{if } \forall i \\ 0 & \text{otherwise} \end{cases} \quad (2)$$

And $h_i \equiv H$ constant magnetic field.

Generally we have the following conditions

→ Boundary conditions do not matter in the thermodynamic limit.

→ Transfer matrix $\mathcal{T} = \begin{bmatrix} e^{\beta H + \beta J} & e^{-\beta J} \\ e^{-\beta J} & e^{-\beta H + \beta J} \end{bmatrix}$ We showed $Z = \text{tr}(\mathcal{T}^N) = \lambda_1^N + \lambda_2^N$ where $\lambda_{1,2}$ are eigenvalues of $\mathcal{T}$.

$$\lambda_{1,2} = e^{\beta J} \left(\cosh(\beta H) \pm \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right) \quad (3)$$

In the thermodynamic limit assuming $\lambda_1 > \lambda_2$

$$\implies \mathcal{F} = -kTN \ln \lambda_1 \quad (4)$$

Assume that thermodynamic potential e.g $\mathcal{F}$ exists in the thermodynamic limit,

$$\mathcal{F}(g_i), \{g_i\} \equiv \text{couplings} \quad (5)$$

$\mathcal{F}$ is analytic except at some points, lines, hyper surfaces in the space of couplings $\{g_i\}$.

Then we can define the 'phase boundary' as a co-dimension one surface in coupling space where $\mathcal{F}$ is not analytic (non analytic surface is the dimension of the space minus one).

It can be shown that $\mathcal{F}$ is continuous

$$\frac{\partial \mathcal{F}}{\partial g^i} \text{ discontinuous} \implies \text{1st order p.t} \quad (6)$$

$$\frac{\partial \mathcal{F}}{\partial g^i} \text{ not discontinuous} \implies \text{continuous p.t.} \quad (7)$$

Properties of $\mathcal{F}$ in 1D- ising

$$\mathcal{F} < 0, \text{continuous} \quad (8)$$

$$\frac{\partial \mathcal{F}}{\partial g^i} \text{ exists almost everywhere} \quad (9)$$

Left-right derivatives exist and are equivalent everywhere

$$S = -\frac{\partial \mathcal{F}}{\partial T} \geq 0 \quad (10)$$

$$\frac{\partial^2 \mathcal{F}}{\partial T^2} \leftrightarrow C_V \geq 0 \quad (11)$$

$$\frac{\partial^2 \mathcal{F}}{\partial H^2} \leq 0 \leftrightarrow \chi_T \geq 0 \quad (12)$$

Are these transitions possible i.e., are there regions of non-analytic city of $\mathcal{F}$?

Recall

$$\mathcal{F} = -kTN \ln \lambda_1, \lambda_1 > \lambda_2 \quad (13)$$

$\mathcal{F}$ might be non-analytic iff

λ_1 is not analytic

$\lambda_1 = \lambda_2$ or

$\lambda_1 = 0$

Perron's theorem

A nxn matrix $\mathcal{A}$ with $\mathcal{A}_{i,j} > 0 \forall i, j$ then $\exists \lambda_1 > \forall j$ such that

$\lambda_1 > 0$

non- degenerate ($\lambda_1 \neq \lambda_2$)

λ_1 is analytic function of $\mathcal{A}_{ij}$

These criteria rule out the possibility for non-analytic behavior of $\mathcal{F} \implies$ no phase transition. This is true for $T > 0$.

We are interested in the regime where the theorem may not be applicable in $T = 0$ or $\beta \rightarrow \infty$ or $J \rightarrow \infty, \beta J \gg 1$

$$\lambda_1 = e^{\beta J} \left(\cosh(\beta H) + \sqrt{\sinh^2(\beta H) + e^{-4\beta J}} \right) \quad (14)$$

for $\beta J \gg 1$

$$\approx e^{\beta J} (\cosh(\beta H) + \sinh(\beta H)) \quad (15)$$

$$\rightarrow \lambda_1 \approx e^{\beta J + |\beta H|} \quad (16)$$

$$\mathcal{F} = -kNT \ln \lambda_1 \approx -N(J + |H|) \quad (17)$$

$$\langle M \rangle = -\frac{1}{N} \frac{\partial \mathcal{F}}{\partial H} = \begin{cases} 1 & \text{if } h > 0 \\ -1 & \text{if } h < 0 \end{cases} \quad (18)$$

Discontinuity in the derivative of $\mathcal{F}$! $\implies$ p.t at $T = 0$!

we want to consider correlations between our thermodynamic quantities.

$$\langle s_i s_j \rangle \leftrightarrow \text{discrete analog of two-point correlators in QFT (or propagators)} \quad (19)$$

j here isn't necessarily $i \pm 1$ rather it can be arbitrary.

We define the correlation functions or two-point functions

$$G_{ij} \equiv G(i, j) = \langle s_i s_j \rangle - \langle s_i \rangle \langle s_j \rangle \quad (20)$$

$$= \langle (s_i - \langle s_i \rangle) (s_j - \langle s_j \rangle) \rangle \quad (21)$$

$$= \langle s_i s_j \rangle \quad (22)$$

because here $\langle s_i \rangle = 0 \forall i = 1, 2, \dots, N$

$G_{i,j}$ is also related to the probability $Pr(s_i = s_j)$

$$Pr_{ij} = \frac{1}{Z} \sum_{ij} \delta_{s_i, s_j} e^{-\beta H} = \langle \delta_{s_i s_j} \rangle = \left\langle \frac{1}{2} (1 + s_i s_j) \right\rangle \quad (23)$$

$$\implies Pr_{ij} = \frac{1}{2} + \frac{1}{2} G_{ij} \quad (24)$$

How to compute G_{ij}

Set $H \neq 0$

first compute $G_{i,i+1}$

$$G(i, i+1) = \frac{1}{Z} \sum_{s_1, s_2, \dots, s_N} s_i s_{i+1} e^{\beta J s_1 s_2 + \dots + \beta J s_{N-1} s_N} \quad (25)$$

$$(26)$$

we want to bring down some part so we take derivative

$$= \frac{1}{Z} \frac{\partial}{\partial (\beta J_i)} \left(\sum_{s_1, \dots, s_N} e^{\beta J_1 s_1 s_2 + \dots + \beta J_{N-1} s_{N-1} s_N} \right) \quad (27)$$

$$G(i, i+1) = \frac{\partial}{\partial (\beta J_i)} \ln Z(J_{i,j} \beta) \quad (28)$$